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A multi-period capacitated facility location problem with maximum travel time and backup service for locating and sizing EMS stations

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Abstract

Emergency medical services (EMS) is a system that provides emergency medical care for incidents involving serious illness or injury. The location of EMS stations plays an essential role in delivering effective and efficient medical services. Numerous location models have been developed for locating and sizing EMS stations. However, it remains challenging to satisfy all EMS planning criteria within a single location model. In this study, a multi-period capacitated facility location problem with maximum travel time and backup service (EMSLSP) is proposed for locating and sizing EMS stations. The most important criteria for EMS planning are taken into account in EMSLSP: the demand changes due to population mobility, the maximum service capacity of an ambulance, the maximum number of ambulances at each EMS station, the maximum travel time from each EMS station to the locations it serves, the full coverage of dynamic demand, the minimum percent of population covered by EMS service in a specific travel time, and a backup station for each demand location in case of need. A case study in Zhengzhou, a large city in China, demonstrates that effective and efficient locations and sizes of EMS stations can be determined by solving the EMSLSP with various planning parameters. Compared with the existing EMS systems, the average ambulance travel time and the percentage of the population served are significantly improved. Simulations of ambulance scheduling confirm that the relocated and resized EMS stations perform better than those in the existing system. The evaluation-optimization-simulation method outlined in this paper provides a comprehensive and effective approach for EMS station planning.

Keywords Emergency medical services, Dynamic demand, Ambulance capacity, Backup service, Location problem, Case study

1 Introduction

Emergency medical services (EMS), an important part of any health service system, provides urgent pre-hospital treatment and stabilization for serious illness and injuries. The response time is the most essential

measure for assessing EMS performance from the citizen's perspective. However, the EMS response times differ from country to country, leading to varying service outcomes. In Brazil, the average response time was 14 min 25 s in 2018 (Colla et al., 2023). In Vienna, Austria, the average response time was 15 min in 2015 (Cabral et al., 2018). In Shanghai, China, the average response time was less than 12 min in 2020 (<http://service.shanghai.gov.cn>). In the United States, the National Fire Protection Association's Standard 1710 has established a 5 min first response time benchmark goal for not less than 90% of dispatched incidents (<https://www.nfpa.org>). In the United Kingdom, it should be a

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maximum of 8 min for 75% of urgent calls and 19 min for serious, but not urgent calls (Bevan & Hamblin, 2009). According to the World Health Organization (WHO), an ideal response time is less than 8 min (Nogueira et al., 2016).

The locations and sizes of EMS stations play an essential role in delivering effective and efficient medical services. First, EMS response time is closely related to the location of ambulance stations and ambulance dispatching (Goldberg & Paz, 1991). The response time mainly depends on the distances from patients to their nearest EMS stations. Second, the response time will also be guaranteed by the sufficient ambulances at each station, especially in the densely populated regions. Due to high population mobility and the dynamic EMS demand in large cities, the number of ambulances required at each station may vary from time to time. A large number of location models have been developed for locating and/or sizing EMS stations. Nevertheless, it remains challenging to satisfy all EMS planning criteria within a single location model.

In this paper, a new mathematical model is proposed for locating and sizing EMS stations. It is a multi-period capacitated facility location problem with maximum travel time and backup service (EMSLSP). Different from the existing models, the proposed model takes into account the most important EMS planning criteria. The criteria are listed as follows: the demand changes due to population mobility, the maximum service capacity of an ambulance, the maximum number of ambulances that can be hosted at each EMS station, the number of ambulances assigned to each EMS station in each time period, the maximum travel time from each EMS station to the locations it serves, the full coverage of dynamic EMS demand, the minimum percent of population covered by EMS service in a specific travel time, and the secondary station assigned to each demand location for backup service. The objective of EMSLSP is to minimize total costs of EMS stations, ambulances, and crews while satisfying the above criteria. The effectiveness of EMSLSP is verified by a case study in Zhengzhou, a large city in China.

This study makes two significant contributions to EMS planning. First, compared with existing models, the proposed model integrates more EMS planning criteria. As a result, the model solution is more effective in terms of emergency response time, system cost, and service reliability. Second, the locations of EMS stations and the number of ambulances required at each station in each time period can be determined simultaneously. Based on the time-dependent model results, ambulance and crew resources can be efficiently scheduled, and the utilization rate of EMS resources can be further improved, thus reducing the total cost of the EMS system.

The study is organized as follows. EMS station location planning and ambulance relocating are briefly reviewed in Sect. 2. A mixed integer linear programming (MILP) model for EMSLSP is formulated in Sect. 3, and verified by a case study in Sect. 4. Several concluding remarks and future research directions are presented in Sect. 5.

2 Literature review

Classical location models have been widely used for locating EMS stations. Among these models, the location set covering problem (LSCP) was first proposed to find the minimum number of emergency service facilities to cover all demands in a predefined service radius (Toregas et al., 1971). Since full coverage is hard to achieve in reality due to limited resources, the maximal coverage location problem (MCLP) (Church & Reville, 1974), which maximizes the demand coverage of a number of facilities with a service radius, has been popularly used in site selection, e.g. Eaton et al. (1979), Eaton et al. (1985), Eaton et al. (1986), and Paulican and Ortega (2013). The p -median problem (PMP), proposed by Hakimi (1964), has also been used to select the location of EMS facilities in Carbondale in the Los Angeles area (Jia et al., 2007; Paluzzi, 2004).

A notable improvement to the classical location models is to redefine the term “demand coverage”. Hogan and Reville (1986) proposed two models, in which the demand points are covered by two facilities (BACOP1) or three facilities (BACOP2). The double standard model (DSM), proposed by Gendreau et al. (1997), used two distance standards r_1 and r_2 ($r_1 < r_2$) to cover the demand. While all the demands must be covered within r_2 , its objective was to maximize the amount of covered demands within r_1 . The backup double covering model (BDCM), formulated by Başar et al. (2009), maximized the coverage of total population using two different EMS stations with different predefined distances. Karatas and Yakıcı (2019) added the criterion of backup service to the PMP, and evaluated the effects of backup service using simulated discrete events.

Another important issue is the service capacity of EMS stations. Pirkul and Schilling (1989) developed the capacitated version of the coverage backup model. Shiah and Chen (2007) introduced ambulance service capacity into the LSCP. This limits the service area of an ambulance, and also avoids work overload. Yin and Mu (2012) proposed capacitated MCLP, which ensures that the demand assigned to each facility does not exceed its capacity. Tozan et al. (2015) also used the capacity LSCP and MCLP to locate the EMS stations in Pendik, Istanbul. Li et al. (2022) used MCLP to select new EMS stations, then optimized the station capacities by a quadratic programming (QP) of spatial accessibility in Shanghai, China.

Jánošíková et al. (2015) proposed a capacitated PMP to locate workload-balanced EMS stations.

Different from the classical location models and their variants, stochastic location models have also been proposed to explicitly optimize the location of EMS facilities with maximal service reliability. The probabilistic LSCP and the queuing probabilistic LSCP (Revelle & Hogan, 1989a; Ball & Lin, 1993; Marianov & Revelle, 1994) consider that the incident locations are random, and covered by EMS stations with probability. Daskin (1983) modified the MCLP to the maximal expected coverage location problem (MEXCLP) with the assumption that the ambulances have the same probabilities of being busy. The maximum availability location problem (MALP) assumes that busy probabilities of vehicles depend on the number of available ambulances and the overall demand (Revelle & Hogan, 1989b; Brotcorne et al., 2003). Karasakal & Karasakal (2004) developed a gradual coverage version of MCLP, which models the gradual decline of coverage as the distance increases. Based on distance decay functions, Berman et al. (2010) formulated the cooperative LSCP (CLSCP) and the cooperative MCLP (CMCLP) models. The former model could decrease the number of required facilities; the latter may increase the demand coverage.

The uncertainties of demand and travel time are also introduced to improve the effectiveness of EMS. Goldberg and Paz (1991) constructed a random model assuming a normal distribution of travel time. Zhang et al. (2017) used the Monte Carlo method to simulate the number of calls and constructed an expected stochastic programming model considering the randomness of demand space. Yang et al. (2020) established a chance constrained programming model considering the randomness of demand space. These stochastic problems are usually transformed into deterministic programming problem to solve (Murali et al., 2012; Zhang et al., 2014).

Additionally, models for ambulance dispatching have been proposed to meet the dynamic service demand. In some cities, the ambulance dispatchers may constantly re-deploy ambulances to busier areas for quicker emergency response. Repede and Bernardo (1994) proposed a dynamic model (TIMEXCLP) considering the changes of emergency demand, driving time and number of ambulances at different periods. Gendreau et al. (2001) proposed a dynamic model (RP^t) for real-time ambulance relocation. The dynamically available coverage location (DACL), proposed by Rajagopalan et al. (2008), is used for dynamically redeploying ambulances to time-varied demands. Zarandi et al. (2013) used the dynamic-demand MCLP to solve large-scale EMS location problems.

Even if various location models have been developed for locating EMS stations, it is still challenging to satisfy the most important EMS planning criteria using the

existing models. First of all, the population mobility is not considered in classical location models or their variants for EMS. Urban studies in recent years show that the population mobility in cities, especially in the large cities, has been boosted extensively by economic growth and modern transport systems. As a result, the EMS demand dynamics should be taken into consideration in providing EMS. Second, in the stochastic models, how to quantitate the uncertainty of EMS needs, the uncertainty of travel time, and the randomness of emergency location have always been challenging tasks (Aringhieri et al., 2016; Bélanger et al., 2018). The uncertainty models for EMS fail in accurately representing and addressing unpredictable factors, which can substantially impact the reliability and robustness of the EMS system. In addition, the models that consider the movement of the crowd and the dynamic change of demands are difficult to solve, and most of them are used in small case studies (Aringhieri et al., 2016; Li et al., 2021). Consequently, it seems very difficult to use these stochastic models for real world EMS planning. Finally, the site-selection of EMS stations and the deployment/redeployment of ambulances at stations were usually solved separately. Locating and sizing EMS stations in a simultaneous manner can further reduce the service cost in that a smaller fleet of ambulances is needed.

In this study, a new model (EMSLSP) is proposed for locating and sizing the EMS stations. We overcome the limitations of existing models by formulating a deterministic model with multi-period demands. In EMSLSP, the dynamic EMS demand at each location is simplified as a set of time-dependent demands. The model is deterministic and thus is relatively easier to solve than the stochastic models. In addition, the locating and sizing problems of EMS stations are simultaneously optimized to save costs on ambulances and human resources.

3 Problem formulation

In this section, a new mathematical model is formulated for locating and sizing EMS stations. For EMS systems, five optimization problems are critical to delivering effective and efficient service: the location problem for site-selection of EMS stations; the sizing problem for determining the station sizes in terms of the number of ambulances; the ambulance relocating problem for quick response to dynamic emergency calls; the EMS crew scheduling problem for effective use of human resources; and the real-time ambulance dispatching problem. The locating and sizing problems are strategic; the ambulance relocating and crew scheduling problems are tactical; and the ambulance dispatching problem is operational. This research focuses on the two strategic problems, and considers both simultaneously. The newly proposed

problem aims to determine the locations and sizes of EMS stations with minimum costs such as the fixed cost of EMS stations, the fixed cost of ambulances, the fixed cost of human resources, and the travel cost of ambulances, while satisfying the constraints such as the maximum travel time of an ambulance, the ambulance service capacity, and the service coverage of an EMS station. The response time, demand coverage and service cost are the key performance indicators for locating EMS stations. To improve these indicators, several critical issues must be addressed.

- (1) Ambulance travel time. The EMS response time is the time from when the emergency center receives the call to when the ambulance arrives at the demand location. It consists of the “turnout time” and the “travel time”. The “turnout time” is often related to the workflow of emergency centers, and it is usually short (Yang et al., 2020). The response time mainly comprises of the ambulance travel time; therefore, it is reasonable to use the travel time instead of the response time in location planning (Rider, 1976; Chelst et al., 1979). Statistics on ambulance travel times, such as the maximum travel time, the average travel time, and the covered demand in a certain travel time, are useful to evaluate the performance of an EMS system. At the same time, the maximum travel time, the travel time with a guarantee percentage of covered demand and other time indicators, can be formulated as model constraints for locating EMS stations that satisfy some planning requirements.
- (2) Maximum service capacity. The potential demand assigned to each EMS station or ambulance should not exceed its service capacity. Ambulance capacity can be defined as the maximum population that can be served by one ambulance, or the maximum number of dispatches per ambulance per day. The capacity of an EMS station is the total capacity provided by ambulances hosted at the station. The capacity constraints on ambulances ensure their availability for each emergency call, most of the time.
- (3) Dynamic demand. The EMS demand depends on the population distribution, age structure and population mobility, and it varies over time due to temporal patterns of different incidents (Degel et al., 2015). Spatiotemporal variation in EMS demand is one of the most challenging issues in EMS location planning. Ambulance relocating among EMS stations is widely used to meet the dynamic EMS demand. In this study, the dynamic EMS demand at

each location is simplified as a set of time-dependent demands.

- (4) Demand coverage. As a public service, EMS should cover 100% of demand spatiotemporally. In EMS planning, one goal is for each demand location to be served by ambulances from an EMS station within a predefined service level, specifically regarding maximum ambulance travel time. Decision makers usually use a demand coverage rate, which is the minimum percentage of the population covered by EMS stations within a specific travel time. For example, the EMS decision makers may hope that 80% of the demand can be served in 10 minutes. This minimum percentage can be used as a planning parameter of an EMS system.
- (5) Backup coverage. Backup coverage means having extra ambulance available when the demand is unusually high. This coverage is essential, especially in high demand areas (Araz et al., 2007). Spatiotemporal variation in EMS demand is each demand location is covered by two EMS stations, a primary station and a secondary station.
- (6) The number of ambulances at EMS stations. The number of ambulances at each station can be given according to station site conditions. It can also be defined as decision variables. The emergency demand varies dynamically, thus the number of ambulances required at each station maybe different in different time periods. Based on the dynamic demand for ambulances at EMS stations, the ambulance fleet and human resources can be effectively scheduled.

Considering the above issues, the following notations are introduced as variables in a new location model for locating and sizing EMS stations.

I : set of demand locations;
 J_1 : set of existing EMS stations;
 J : set of candidate EMS stations, and $J_1 \subset J$;
 K : set of daily time periods;
 d_{ik} : population in demand location i during time period k ;
 f_j : fixed cost of EMS station at location j ;
 C_k : operating cost of each ambulance and its crews during time period k ;
 t_{ijk} : travel time from station j to demand location i during time period k ;
 β : traffic cost per distance/time unit;
 α_k : percentage of EMS demand during time period k ;
 t_p, t_b : maximum travel time from primary station and backup station to demand location, respectively, and usually $t_p \leq t_b$;

t_{pr}, φ : minimum percentage of population covered by EMS stations in the travel time t_{pr} , and $t_{pr} < t_p$;

M_j : maximum number of ambulances that can be hosted at EMS station j ;

w_k : maximum population that can be served by one ambulance during time period k ;

P : minimum number of existing stations that must be reserved, $P \leq |J_1|$.

The decision variables are defined as follows:

x_{ijk}^p : if demand point i is served by primary station j during time period k , $x_{ijk}^p=1$, otherwise $x_{ijk}^p=0$;

x_{ijk}^b : if demand point i is served by backup station j during time period k , $x_{ijk}^b=1$, otherwise $x_{ijk}^b=0$;

y_j : if candidate location j is selected, $y_j = 1$, otherwise $y_j = 0$;

V_{jk} : number of ambulances required at station j during time period k .

A mixed integer linear programming model for locating and sizing EMS stations can be formulated as follows:

$$\text{Minimize : } \sum_{j \in J} f_j y_j + \sum_{j \in J} \sum_{k \in K} C_k V_{jk} + \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} \beta \alpha_k d_{ik} t_{ijk} x_{ijk}^p \quad (1)$$

$$\text{Subject to : } \sum_{j \in J, t_{ijk} \leq t_p} x_{ijk}^p = 1, \forall i \in I, k \in K \quad (2)$$

$$\sum_{j \in J, t_{ijk} \leq t_b} x_{ijk}^b = 1, \forall i \in I, k \in K \quad (3)$$

$$\sum_{i \in I} d_{ik} \sum_{j \in J, t_{ijk} \leq t_{pr}} x_{ijk}^p \geq \varphi \sum_{i \in I} d_{ik}, \forall k \in K \quad (4)$$

$$\sum_{i \in I} d_{ik} x_{ijk}^p \leq w_k V_{jk}, \forall j \in J, k \in K \quad (5)$$

$$x_{ijk}^p + x_{ijk}^b \leq 1, \forall i \in I, j \in J, k \in K \quad (6)$$

$$x_{ijk}^p \leq V_{jk}, \forall i \in I, j \in J, k \in K \quad (7)$$

$$x_{ijk}^b \leq V_{jk}, \forall i \in I, j \in J, k \in K \quad (8)$$

$$\sum_{j \in J_1} y_j \geq P \quad (9)$$

$$V_{jk} \leq M_j y_j, \forall j \in J, k \in K \quad (10)$$

$$x_{ijk}^p, x_{ijk}^b \in \{0, 1\}, \forall i \in I, j \in J, k \in K \quad (11)$$

$$y_j \in \{0, 1\}, \forall j \in J \quad (12)$$

$$V_{jk} \in \{0, 1, 2, 3, \dots\}, \forall j \in J, k \in K \quad (13)$$

The objective function (1) minimizes the total EMS cost including the fixed cost of EMS stations, the fixed cost of ambulances and EMS crews, and the travel cost of ambulances. Constraint (2) ensures that each demand point i is covered by a primary station at time period k in a maximum travel time t_p . Similarly, constraint (3) ensures that each demand point i is covered by a backup station at time period k for a maximum travel time t_b . Constraint (4) guarantees that the primary stations cover φ of population in travel time t_{pr} at time period k . The variable t_{pr} and φ are often used as EMS planning parameters. Constraint (5) specifies that the demand assigned to each station j in time period k does not exceed the capacity of ambulances at station j . Constraint (6) means that the primary station and

backup station for each demand i must be different during time period k . Constraints (7) and (8) ensure that the demand i can only be served by station j as a primary or backup station when one or more ambulances are hosted at station j at time period k . Constraint (9) states that at least P existing stations in set J_1 must be reserved. Constraint (10) is that the number of ambulances at the station cannot exceed its maximum number. Constraints (11), (12) and (13) define the decision variables x_{ijk}^p , x_{ijk}^b , y_j and V_{jk} .

In planning practice, the commonly used constant variables such as the fixed cost of each EMS station (f_j), the fixed cost of an ambulance and its crew professionals (C_k), the ambulance travel times (t_{ijk}), the maximum service capacity of ambulance during time period k (w_k), and the emergency call rate (α_k) can be estimated by historical EMS statistics. The maximum number of ambulances hosted at each station (M_j) can be set according to its site conditions. The locations and sizes of EMS solutions are largely determined by the planning parameters t_p , t_b , t_{pr} , φ and P . These parameters can be set according to the official EMS standards, the professional association's recommendations, or the choice of EMS decision makers.

4 Case study

4.1 Study area

The EMSLSP model was tested in urban Zhengzhou, the capital city of Henan province, and one of the largest cities in central China. In recent years, the urban area of Zhengzhou has gradually expanded from the central region towards the periphery, forming a concentric ring structure. The population mobility in Zhengzhou, as well as other large cities in China, is getting higher and higher as employment opportunities, infrastructure and public services are mainly concentrated in the central areas of the city (Zheng et al., 2020).

In the study area, there are one EMS center (Zhengzhou Emergency Medical Rescue Center), 54 EMS stations and 179 ambulances. The EMS stations are mainly concentrated in the central areas, and most are located in hospitals (Fig. 1). According to the Center's annual report, in 2021, Zhengzhou received 1.0 million calls, of which 0.62 million were received in the urban area. All the ambulances are dispatched in real-time by the Center. According to our interview with the administrative staff of the Center, conducted in November 2023, determining the optimal locations for EMS stations to cost-efficiently reduce service response time is one of the most challenging issues.

The geographic data for EMS station planning were prepared in ArcGIS 10.3. The urban area of Zhengzhou, shown in Fig. 1, is 566.43 km². It was divided into 2375 grids (500 m × 500 m), and the center point of each grid was considered as a demand point (Fig. 1). 411 locations of existing EMS stations, fire stations, hospitals and community-level healthcare centers were selected as candidate stations (Fig. 1). The maximum number of ambulances at each existing station was set according to its existing ambulances. The maximum number of ambulances at each new candidate station was set according to its site conditions. We set 8:00–20:00 as the daytime, and 20:00 to 8:00 next day as the nighttime. The distances from stations to demand locations were estimated by Manhattan distance. The average speeds of ambulance were set to 15 km/h in daytime, and 20 km/h at nighttime.

4.2 Population

The dynamic population in each grid was estimated by the following procedures. The building data in the study area were downloaded from Rivermap (<http://www.rivermap.cn/>). We labeled each building as residential or non-residential (Fig. 1). The population in each building is calculated as $P_i = F_i * A_i / A$, where F_i is the number of

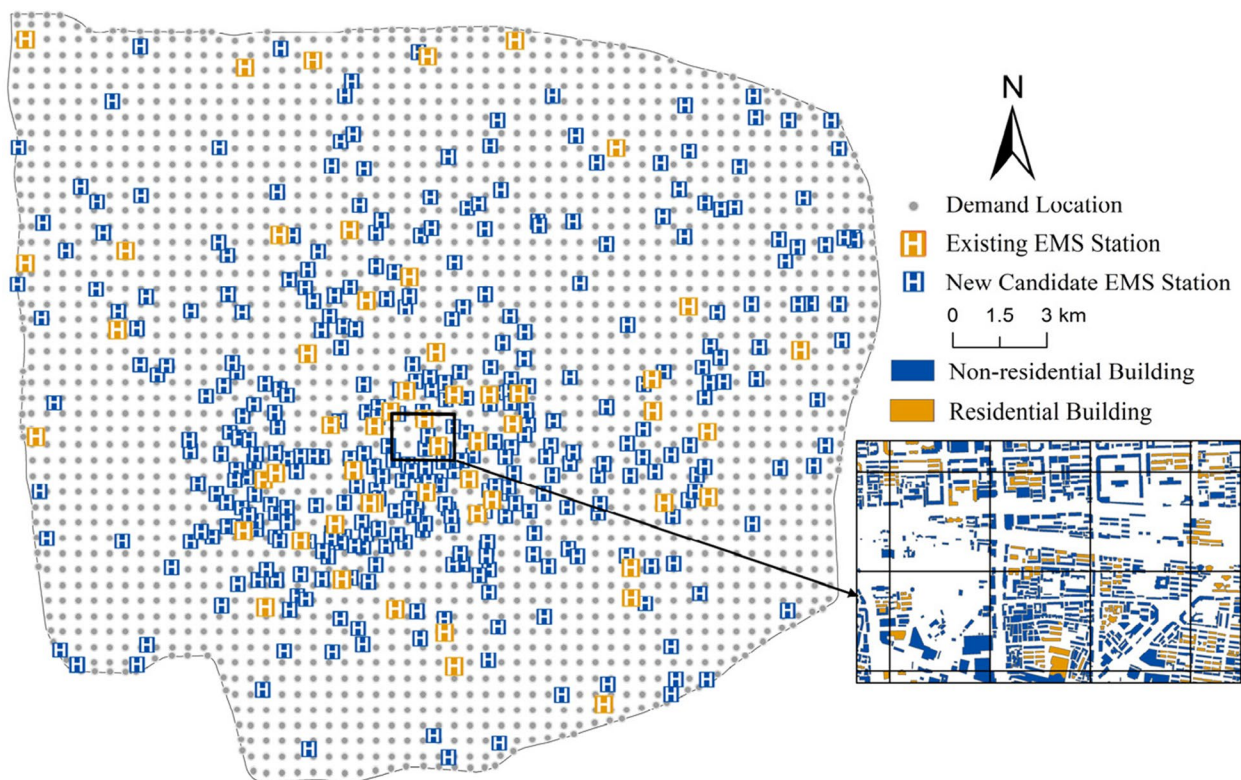


Fig. 1 Map of the study area

floors of the building i , A_i is the base area of building i , and A is the per capita housing area. The population in each grid is estimated by summarizing the population inside the grid. The population in a boundary building is split as $P_{ij} = P_i * B_j / A_i$, where B_j is the area of building i in grid j (Li et al., 2022). The residential population is considered as nighttime population, which is estimated by using residential buildings. The daytime population is estimated by using residential and non-residential buildings (Li et al., 2022). In the study area, the populations were 5.32 million in daytime and 5.11 million at nighttime, as shown in Fig. 2.

4.3 Spatial accessibility of existing EMS stations

To evaluate the spatial accessibility of existing EMS stations, the ambulance travel time to each demand location is estimated by the optimal supply–demand (OSD) model (Zhai et al., 2022). The service supply stations are optimally assigned to demand locations with minimum travel cost while satisfying their maximum capacities. Table 1 shows several statistics on the ambulance travel time. The average travel time from EMS stations to their served locations is 11.26 min in daytime, which is 3.36 min longer than that at nighttime. In the two time periods, 74.81% and 86.80% of population can be covered by EMS stations in 10 min of travel time, respectively.

4.4 Locating and sizing EMS stations

The EMSLSP model in Sect. 3 was used for locating and sizing EMS station in the study area. The constant variables and the cost-related parameters used in the model were set according to EMS statistics in China. In the

Table 1 Statistics on the estimated ambulance travel times in existing EMS system

Time period	Average travel time (m)	Maximum travel time (m)	Population covered ≤ 5 m (%)	Population covered ≤ 10 m (%)	Population covered ≤ 15 m (%)
Daytime	11.26	38.72	38.21	74.81	90.22
Night-time	7.90	28.13	48.73	86.80	97.08

case study, the cost parameters in objective function (1) are normalized on a daily basis and further divided into two time periods: daytime and nighttime. The fixed daily cost of each EMS station, f_j , is set to ¥410.96. The daily fixed costs of each ambulance and its crews is ¥1095.89 (Su et al., 2015). In this study, the costs of each ambulance and its crews during both daytime and nighttime are assumed to be same; as a result, both the parameters C_d and C_n during the daytime and nighttime are set to ¥547.945. Parameters α_d (daytime) and α_n (nighttime) are set to 0.004% and 0.003% of the total population, respectively (Liu et al., 2021). The parameter β is set to ¥1.0 per minute of travel time. On September 17, 2020, the National Health Commission of the People's Republic of China recommended that one ambulance serves 30,000 people in a prefecture-level city (National Health Commission of the People's Republic of China, 2020). Therefore, the service capacity of each ambulance at daytime, w_d , is set to a population of 30,000. Since the emergency requests during the daytime is more than that during the nighttime (Liu et al., 2021), the parameter w_n at nighttime is set to a population of 40,000.

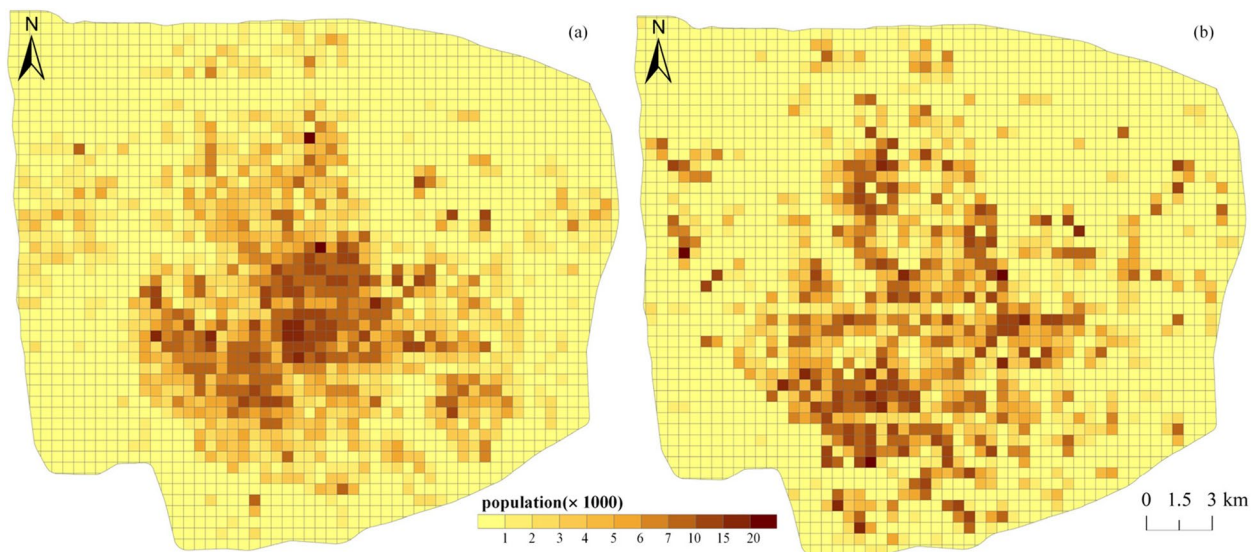


Fig. 2 Population distribution in daytime (a) and at nighttime (b)

The planning parameters are set as follows. The maximum travel times, t_p and t_b , are set to 15 min and 18 min, respectively. The minimum demand coverage parameters, t_{pr} and φ , are set to 10 min of travel time and 80% of total population, respectively.

Three planning scenarios are designed in this case study: whole new planning, incremental planning and adjusting planning. For whole new planning ($P=0$), all the existing stations are considered as candidate locations. For incremental planning ($P=54$), all existing stations must be reserved in the final solution and some locations will be selected from candidate locations for setting new EMS stations. For adjusting planning ($0 < P < 54$), at least P existing stations must be reserved. As a result, some existing stations will be closed. In this study, the parameter P is set to 0, 30, 40, 50 and 54.

The model instances with different sets of planning parameters are generated by Python programming, and solved by the state-of-the-art optimization solver Gurobi 9.0.3. It is well known that the capacitated facility location problem (CFLP) is computationally NP-hard. The EMSLSP, which is formulated based on the single-source CFLP (SSCFLP), is even more complex to compute. However, it was fortunate that the model instances for the study area were solved near-optimally on a desktop computer with single Inter Core i5-10500 CPU, and 16 GB RAM.

Three model solutions are shown in Fig. 3. For the model with parameter $P=0$, the EMS locations and the number of ambulances at each station are shown in Fig. 3(a). The stations are presented as solid H (existing stations) or hollow H (new stations). Among the 71 EMS stations, 23 are the existing ones. The numbers of ambulances required at each EMS station in daytime and nighttime are also allocated. The numbers of ambulance are illustrated above the station symbols. Based on the optimized stations, every demand location can be reached in 15 min by an ambulance from its primary station, or 18 min by an ambulance from its secondary station. The average travel times from primary stations to their customers are 7.29 min in daytime and 5.58 min at nighttime. Compared with the existing EMS system, the two travel times are reduced substantially. The model results from $P=30$ and $P=54$ are shown in Fig. 3(b) and (c), respectively.

All the solution results from the study area are summarized in Table 2. In the Table, column T shows the computation times of Gurobi optimizer in hours; the columns Cost and Gap show the objectives and optimality gaps of the solutions, respectively. The number of EMS stations is shown in columns M. The total number of ambulances required, the average and maximum ambulance travel times are shown in columns N1, T1, and MT1,

respectively. The three indicators at nighttime are presented in columns N2, T2, and MT2, respectively.

There are at least three important findings from the planning results. First, all the planning criteria can be satisfied. Every demand location is served by its primary station in 15 min of travel time, and by its backup station in 18 min of travel time. The total population served by each EMS station at each time period is not greater than its service capacity. Second, the more existing stations are kept, the greater the number of stations and operation costs are required to satisfy the planning criteria. Third, compared with the existing EMS system, the average travel times of ambulances are decreased substantially, at least 34.81% in daytime and 27.59% at nighttime. The maximum travel times from primary stations are 15.00 and 14.97 min during the daytime and nighttime, respectively. The population covered by stations in 10 min is much higher than that of existing stations. The number of ambulances is significantly higher than emergency stations. Therefore, the cost of an EMS systems is dominated by the cost of ambulances and crews, and the cost of ambulances and crews > the cost of EMS stations > the total travel cost of ambulances.

In realistic planning, most existing EMS stations are usually kept to save capital resources. Figure 3 shows that the EMS station locations change significantly when different scenarios are set for parameter P . The numbers of ambulances required at each station in different periods are also different. Moreover, the number of ambulances required in the central area is significantly greater than that in the peripheral regions. The number of ambulances required at each EMS station during the daytime is significantly different from that during the nighttime. The detailed numbers of ambulances at each station are showed in Tables S1-S5.

The percentages of population covered by EMS stations in different travel times are summarized in Table 3. Using the optimized EMS stations, all the population in the study area can be served by ambulances in 15 min of travel time. Compared with existing stations, the percentages of population covered by stations in 10 min of travel time are increased from 74.92% to 82.96–88.21% in daytime and from 86.80% to 95.35–96.85% at nighttime. In all model results, more than 80% of population can be served in 10 min of travel time, which is guaranteed by two planning parameters $t_{pr}=10$ min and $\varphi=80\%$.

4.5 Sensitivity of planning parameters

The model solutions are closely related to the planning parameters. The sensitivity of parameters t_p , t_b , t_{pr} , φ , w_d and w_n can be analyzed by controlled experiments. First, an adjustment planning scenario ($P = 50$) with

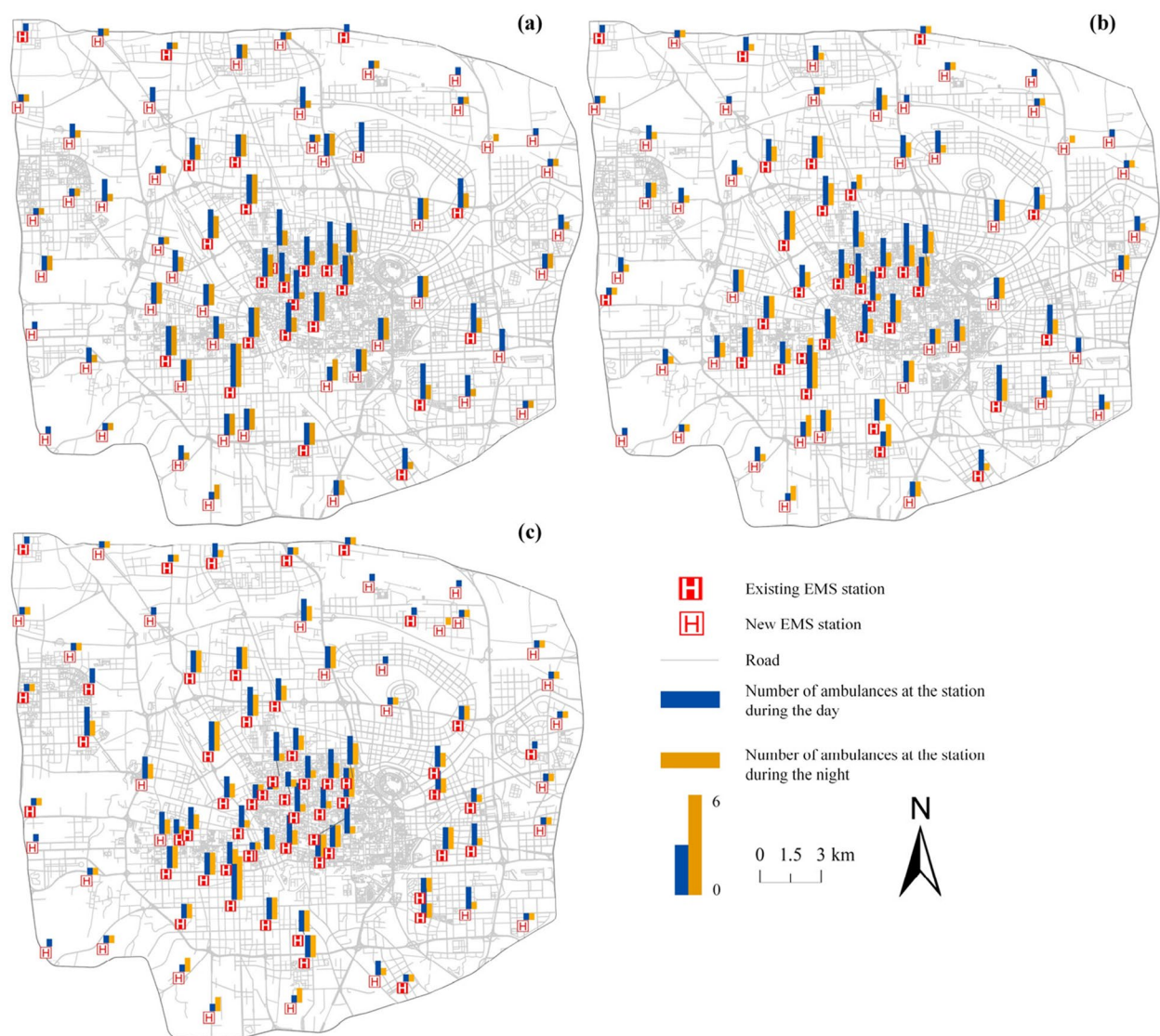


Fig. 3 Locations and sizes of EMS stations: (a) whole new scenario ($P=0$), (b) adjusting scenario ($P=30$), and (c) incremental scenario ($P=54$)

Table 2 Solution results from the study area

Planning scenario	Cost	T (h)	Gap (%)	M	N1	T1 (m)	MT1 (m)	N2	T2 (m)	MT2 (m)
Whole new ($P=0$)	188497.7	3.80	2.00	71	183	7.29	14.99	130	5.58	14.97
Adjusting ($P=30$)	188366.3	3.84	1.91	72	182	7.34	15.00	130	5.52	14.97
Adjusting ($P=40$)	189383.9	1.41	1.98	73	183	7.27	14.98	130	5.72	14.97
Adjusting ($P=50$)	192514.0	0.72	1.94	79	183	6.85	14.98	132	5.42	14.97
Incremental ($P=54$)	194150.7	0.56	1.92	83	183	6.76	14.98	132	5.50	14.92

parameters $t_p=15$, $t_b=18$, $t_{pr}=10$, $\varphi=80\%$, $w_d=30,000$, and $w_n=40,000$ is used as the control scenario. Second, the experimental scenarios are designed by setting different values to an experimental parameter. The sensitivity of

the parameters can be shown by comparing the control scenario with the experimental scenarios.

Table 4 shows the solutions with different planning parameters. The column Default shows the solution

Table 3 Percentages of population covered by EMS stations in different travel times

Planning scenario	Daytime			Nighttime		
	≤ 5 m (%)	≤ 10 m (%)	≤ 15 m (%)	≤ 5 m (%)	≤ 10 m (%)	≤ 15 m (%)
Existing	38.32	74.92	90.33	48.73	86.80	97.08
Whole new ($P=0$)	37.82	84.02	100.00	63.69	96.85	100.00
Adjusting ($P=30$)	37.85	83.36	100.00	62.47	96.47	100.00
Adjusting ($P=40$)	39.48	82.96	100.00	54.02	95.35	100.00
Adjusting ($P=50$)	47.33	88.21	100.00	60.79	96.53	100.00
Incremental ($P=54$)	48.38	87.95	100.00	60.18	96.28	100.00

from the control scenario. Other columns illustrate the solutions from the experimental scenarios with different experimental parameters. When the maximum travel times, t_p and t_b , are increased, the total EMS cost will be slightly reduced because fewer EMS stations and ambulances are required. At the same time, the average travel time will be increased considerably. When the maximum capacities of an ambulance, w_d and w_n , are decreased, the total EMS cost dramatically increases because more ambulances and emergency professionals are required. However, the average travel time almost remains the same. When the minimum demand coverage, φ , is increased from 80 to 90%, or the threshold travel time, t_{pr} , is decreased from 10 to 8 min, the total EMS cost increases slightly because more EMS stations, ambulances and professionals are required.

Table 4 shows that the total number of ambulances and emergency crews mainly depends on the service capacity of an ambulance in terms of the maximum population that can be served by an ambulance. The average travel time of ambulances is determined by the two parameters t_p and t_b . The total EMS cost is mainly determined by the ambulances and emergency crews; it is slightly affected by other planning parameters. All planning parameters except φ have negative correlations with the EMS cost, the number of ambulances and emergency crews.

4.6 Simulations of ambulance dispatching

The effectiveness of the optimized locations and sizes of EMS stations was verified by well-designed simulations of ambulance dispatching and emergency service. The simulations were implemented based on a time model of ambulance dispatching and emergency service, as shown in Fig. 4. When the EMS call center receives an emergency call, the EMS operator inquires for patient's location and condition immediately. If the patient requires on-site medical service and/or medical transport, the EMS center will dispatch the nearest ambulance to serve the patient. Having received a request, the ambulance will promptly travel to the scene from its station. After the ambulance arrives at the scene, the emergency professionals diagnose and treat the patient. If the patient must go to hospital, the ambulance will transport the patient from the scene to the nearest hospital, then the ambulance will return to its home base. If the patient does not need to be transported to the hospital, the ambulance will return to its home base station from the scene.

The experiments were conducted in two steps: (1) to generate a daily list of EMS incidents with information such as incident location, time of occurrence, time of on-site treatment, and whether a transfer service is required; and (2) for each of the incidents, to dispatch the nearest available ambulance to the site of incident. In the first step, the number of incidents and the incident attributes

Table 4 Statistics of the solutions from the controlled experiments

Parameters	Default	$t_p=17, t_b=20$	$t_p=20, t_b=23$	$w_d=25000, w_n=35000$	$w_d=35000, w_n=45000$	$t_{pr}=10, \varphi=90\%$	$t_{pr}=8, \varphi=80\%$
Cost	192514.0	190399.4	186007.8	221494.8	174002.6	195664.2	194019.6
M	79	75	68	88	82	83	83
N1	183	182	180	220	159	185	184
T1(m)	6.85	7.39	8.01	6.88	6.72	6.65	6.70
MT1(m)	14.98	16.93	19.79	14.98	14.98	14.98	14.93
N2	132	132	131	152	117	133	133
T2(m)	5.42	5.61	6.13	5.26	5.58	5.46	5.40
MT2(m)	14.97	16.22	18.91	14.94	14.92	14.94	14.97

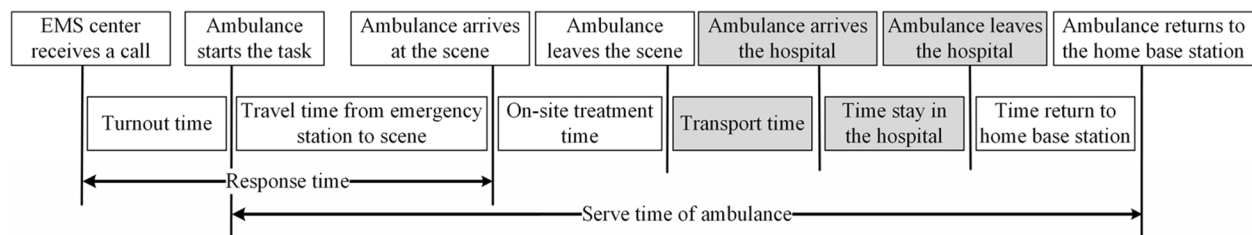


Fig. 4 Time model of emergency service

are randomly set according to the demand locations, time-depended population and historical EMS statistics. In the second step, all the incidents are served sequentially. At the time of occurrence of an incident, the nearest available ambulance will be dispatched to the site of incident and provide needed medical services. The turnout time, travel time, treatment time, transfer time if required, and travel time of return to station are considered for each emergency service incident.

The detailed data used by the simulations were prepared according to the EMS statistics in China. According to Wang et al. (2021) and Ingolfsson (2013), it is assumed that the total number of incidents follows a Poisson distribution. The mean of the daily number of incidents is set according to the historical ambulance dispatching in the study area. We assumed that the daily number of incidents on normal days follows a Poisson distribution with a mean of 372.0. It was also assumed that the daily number of incidents on extremely busy days follows a Poisson distribution with a mean of 1116.0. The roulette wheel method is used to randomly simulate the location of incidents (Golabian et al., 2022). The population in demand locations are accumulated to create a roulette wheel. A demand location on the wheel is randomly selected as the location of each incident. For each incidence, an hourly time period is randomly assigned the incident according to the time statistics on daily EMS incidents; exact time of occurrence is then randomly selected within the time period (Liu et al., 2021). Based on EMS regulations of Zhengzhou, the minimum and average turnout time are

40.0 and 72.0 s, respectively. The time of on-site treatment and stay in hospital obey an exponential distribution $\lambda=0.0834$ and $\lambda=0.2260$, respectively (Ji et al., 2019). According to Yang et al. (2019), 65% patients are admitted to hospital. The ambulances will return to their home stations after completing their tasks. The travel time from EMS station to the site of incident, transfer time from the site of an incident to a hospital and the time of return to station are estimated using the speed of 15 km/h in daytime and 20 km/h at nighttime.

For each planning scenario, the EMS performance on normal days was measured by the time statistics on 365-day daily simulations of ambulance dispatching. The emergency events in each day were prepared randomly and differed from the other days. The simulation results are summarized in Table 5. In the Table, columns T and MT show the average and maximum response times in daytime, respectively. The column, “ ≤ 10 m”, illustrates the percentage of total incidents can be served in 10 min. Compared with the current EMS system, the average response time of the optimized locations is reduced significantly, from 7.79 min to 6.47–6.69 min in daytime, and from 6.52 min to 5.40–5.79 min at nighttime. At the same time, the maximum response time is decreased drastically, from 32.19 min to 15.66–19.98 min in daytime, and from 21.03 min to 15.47–16.41 min at nighttime. Furthermore, 84.0%–86.8% incidents can be serviced in 10 min of response time in daytime, and 95.1%–97.0% at nighttime, which are increased significantly. The backup rate represents the percentage that

Table 5 Simulation results from normal daily events

Planning scenario	Daytime				Nighttime			
	T (m)	MT (m)	≤ 10 m (%)	Backup rate(%)	T (m)	MT (m)	≤ 10 m (%)	Backup rate(%)
Existing	7.79	32.19	73.9	0.52	6.52	21.03	84.5	0.58
Whole new ($P=0$)	6.69	15.66	86.5	0.45	5.43	15.47	96.8	0.73
Adjusting ($P=30$)	6.69	16.13	86.2	0.45	5.40	15.47	97.0	0.74
Adjusting ($P=40$)	6.66	15.83	84.0	0.41	5.79	15.65	95.1	0.80
Adjusting ($P=50$)	6.52	19.10	86.6	0.50	5.50	15.47	96.4	0.93
Incremental($P=54$)	6.47	19.98	86.8	0.48	5.51	16.41	96.1	0.88

the ambulances are dispatched from backup stations. It is less than 0.52% during the daytime, and less than 0.93% during the nighttime.

In addition, 30-day daily events are used to evaluate the EMS performance on extremely busy days. The average number of incidents on extreme days is three times that on normal days. Table 6 summarizes the simulation results from each planning scenario. Compared with the current EMS system, the optimized locations demonstrate a significant decrease in both average and maximum response times. Tables 5 and 6 also show that when EMS demand increases dramatically on extreme days, the average emergency response time increases slightly, less than 1.0%. However, the usage ratio of the ambulances at backup stations increase considerably from 0.41%~0.52% to 2.59~3.79% in daytime and from 0.58%~0.93% to 3.05~5.70% at nighttime.

5 Conclusion

A new model for locating and determining the size of EMS stations, EMSLSP, is proposed in the study. It is formulated as a mixed integer linear programming model. The commonly used criteria for EMS planning are considered in the model. The case study in a large city of Zhengzhou, China, shows that key performance indicators of EMS, such as the maximum ambulance travel time, the average ambulance travel times, and the percentage of population covered by EMS stations in a specific travel time, can be significantly improved by a slightly larger fleet of ambulances. The effectiveness of the proposed model is also verified through extensive simulations. In addition, the evaluation-optimization-simulation method outlined in this paper provides a comprehensive and effective approach for EMS station planning.

The proposed model is different from the existing location models for EMS. First, the most important criteria for locating EMS stations, such as ambulance capacity, dynamic demand, maximum travel time, are formulated

in the EMSLSP as model constraints. For example, the ambulance capacity is not considered in most MCLP-based location problems, such as the MEXCLP, MALP, dynamic-demand MCLP, DSM, BDCM and RP^t . If the required ambulances cannot be hosted at a station due to its physical space, it might be necessary to dispatch ambulances from remote stations (Karatas & Yakıcı, 2019). Since the most importance criteria are taken into account in EMSLSP, it can guarantee service performance and reliability. Second, the locations and sizes of EMS stations can be determined simultaneously to meet dynamic service demand. It is different from the ambulance dispatching models for quicker emergency response, such as TIMEXCLP, RP^t , DACL or two-step optimization methods (Li et al., 2022). Based on the time-dependent results from the EMSLSP, a minimum number of ambulances and crew resources can be efficiently scheduled. This has potentials to improve EMS resource utilization, and thus reduce the total cost of an EMS system. Finally, the locations and sizes of EMS stations are optimized by setting different planning parameters, t_p , t_b , t_{pr} , φ and P . These parameters can be set according to the official service standards, the professional recommendations, and/or the choice of decision makers. Therefore, the EMSLSP can provide a needs-driven, cost-effective and service-reliable approach to locating and determining the size of EMS stations.

When the EMSLSP is defined as a general, flexible, and effective model for EMS planning, several issues need to be addressed for further research. First of all, the model results may be sensitive to the choice of time periods. Since dynamic EMS demands are represented as time-dependent sets, the accuracy of these demands usually vary depending on the number of time periods. Generally, the more time periods used in practice, the more accurately demand data can be generated. For example, the demand estimated in hourly time periods will approximate real-world population mobility. The EMSLSP is capable of handling such hourly demand data. However, this does not mean that 1-h or 10-min time periods are

Table 6 Simulation results from extreme daily events

Planning scenario	Daytime				Nighttime			
	T (m)	MT (m)	≤ 10 m (%)	Backup rate(%)	T (m)	MT (m)	≤ 10 m (%)	Backup rate(%)
Existing	7.90	31.74	73.0	3.79	6.60	23.18	83.5	3.05
Whole new ($P=0$)	6.75	18.70	85.5	2.61	5.54	19.22	95.6	4.26
Adjusting ($P=30$)	6.75	18.53	85.3	2.59	5.52	21.66	95.8	4.35
Adjusting ($P=40$)	6.71	21.42	83.1	2.80	5.72	20.75	94.3	5.24
Adjusting ($P=50$)	6.62	20.77	85.3	3.16	5.63	18.19	94.9	5.57
Incremental($P=54$)	6.57	19.42	85.4	3.14	5.63	18.19	94.5	5.70

recommended for model applications. The availability of data on population mobility should be considered when choosing the time periods. Even if potential inaccuracies in demand estimation and representation could affect the model's reliability in practice, it should still outperform models that do not consider population mobility.

The travel speed of ambulance is another issue in EMS planning, which can significantly impact emergency response time. Ambulance speed depends on multiple factors such as road infrastructure, traffic congestion, and ambulance priority on the road. In our case study, we simply set two speeds for ambulances during daytime and nighttime, respectively. To improve service reliability, it is recommended to prepare the model variables t_{ijk} using a time-dependent traffic network or traffic big data.

The third issue is the design of an efficient and effective heuristic algorithm for the EMSLSP. In our case study, the model instances were near-optimally solved using the Gurobi optimizer in 0.56–3.84 h. For larger EMSLSP instances or those with more time periods, a heuristic algorithm is required. Since the EMSLSP is formulated based on the SSCFLP, the matheuristic algorithm for the SSCFLP (Kong, 2021) could be adapted to solve the EMSLSP. The algorithm starts with an initial solution and iteratively improves it through large neighborhood search, which is implemented by solving sub-problems.

Finally, the EMSLSP should be further tested in other regions. The usefulness of the model largely depends the data preparation. The fixed cost of each EMS station (f_j), the fixed cost of an ambulance and its crew professionals (C_k), the maximum service capacity of ambulance during time period k (w_k), and the emergency call rate (α_k) should be investigated according the history EMS data. The planning parameters t_p , t_b , t_{pr} , φ and P should be set according to service standards. Population mobility and ambulance travel times may be estimated using spatiotemporal big data.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43762-024-00143-z>.

Supplementary Material 1.

Authors' contributions

Guangli Zhang built the model, performed the data processing and analysis, wrote the first draft of the manuscript. Rui Ma performed the estimation of population during the daytime and nighttime. Yunfeng Kong conceptualized the idea and method, improved and revised the manuscript. Chenchen Lian and Hao Guo contributed to investigate the ambulances and EMS stations in Zhengzhou. Shiyang Zhai improved and revised the manuscript.

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Data availability

The data used during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

No conflict of interest exists in the submission of this manuscript.

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